

EE5203 L01 Lab Guide

Build, flash, debug, and modify Blinky - Basri Erdogan

Mission: Create a Nucleo-F401RE project, make LD2 blink, prove that the program executes in the debugger, and independently change its timing.

Prepare before class

- ☐ Complete the five-question **pre-lab quiz** in CATS (**Tests & Quizzes**). Two attempts; the higher score counts.
- ☐ Bring the Question 5 entry ticket.
- ☐ Lab PCs have everything installed: log in with your own student account. On your own laptop, install the tools named in the Reference page first.
- ☐ Bring a USB **data** cable if using your own equipment.

Learning objectives

By checkoff time, you can:

1. build, flash, run, and debug a Nucleo-F401RE project;
2. distinguish build evidence from hardware evidence;
3. set a breakpoint and inspect a variable;
4. make and explain one controlled code modification.

Hardware and files

Item	Required value
Board	Nucleo-F401RE
Programmer/debugger	On-board ST-LINK USB connection
Output	LD2, connected to PA5
Tools	STM32CubeMX (configure/generate) + VS Code (edit/build/debug)
Project name	EE5203_L01_<studentnumber>

Session plan

Time	What happens
0:00 – 0:15	Opening demonstration — one live pass through the complete workflow.
0:15 →	Checkpoints A–D below at your own pace; the oral check happens when you reach D.

The demonstration follows this path once:

```
create -> configure -> generate -> edit -> build -> flash -> debug -> verify
```

Observe which tool owns each stage: the `.ioc` configuration lives in STM32CubeMX; `main.c`, the build terminal, debug controls, and the variable panels live in VS Code. Record questions; do not copy menu clicks without understanding the stage they belong to.

Checkpoint A - Orient: board and tools

1. Connect the board through the ST-LINK USB connector.
2. In **STM32CubeMX**: create a project for **NUCLEO-F401RE** (Board Selector) and accept the board's default peripheral initialization.
3. Confirm that PA5/LD2 is configured as a GPIO output.
4. In Project Manager set the name (no spaces), the location `C:\Work\<studentnumber>`, and **Toolchain/IDE = CMake**; generate code.
5. Copy the folder `C:\Users\Public\EE5203_template\.vscode` into the generated project folder, next to the `.ioc` file (Reference page → Workflow, *Template* row). Without it VS Code cannot build or debug.
6. Open the generated folder in **VS Code** (File → Open Folder → the folder that contains the `.ioc`; choose preset **Debug** if asked) and locate the protected **USER CODE** regions in `main.c`.
7. Build the unchanged project with **Ctrl+Shift+B**: zero errors, and a **Linking C executable** line at the end.

Evidence for Checkpoint A: Show the selected board, PA5 configuration, and a successful build message.

Checkpoint B - Prove: build and flash

Inside the **existing generated while** (1), between `/* USER CODE BEGIN 3 */` and `/* USER CODE END 3 */`, add these statements. Do not create a second infinite loop.

```
HAL_GPIO_TogglePin(LD2_GPIO_Port, LD2_Pin);  
HAL_Delay(500);
```

Build with **Ctrl+Shift+B**, flash with **F5** (Debug (ST-LINK)), press Continue, and verify a stable blink. Explain why the physical observation is stronger evidence than the build message alone.

Checkpoint C - Observe: breakpoint and variable

Add before the loop, between `/* USER CODE BEGIN 2 */` and `/* USER CODE END 2 */`:

```
int toggle_count = 0;
```

Increment it after every toggle:

```
toggle_count = toggle_count + 1;
```

1. Place a breakpoint on or immediately after the increment — and **write down which**: a breakpoint *on* the increment pauses before it runs, so the first value you see is 0; on the line *after* it, the first value is 1. Predict for the line you chose.
2. Start a debug session.
3. Predict the value before pressing **Resume**.
4. Resume twice and record the observed values.

Observation	Predicted	Observed
First pause		
Second pause		
Third pause		

Evidence for Checkpoint C: Show the breakpoint, execution marker, and changing `toggle_count` value.

Checkpoint D - Own: modify and justify

Each student must make one assigned change and demonstrate it individually.

Variant	Requirement
A	Change the delay to 100 ms
B	Change the delay to 250 ms
C	Change the delay to 750 ms
D (stretch)	Blink three times quickly, then pause for 1 s — write the three toggle pairs out; the loop that shortens this arrives in L02

Before flashing, state the predicted behavior. Change only what is necessary, then collect physical or debugger evidence.

Acceptance checklist

- ☐ Correct Nucleo-F401RE project selected.
- ☐ Project builds without errors.
- ☐ LD2 exhibits the required core behavior.
- ☐ Breakpoint is reached and `toggle_count` changes as predicted.
- ☐ Individual timing variant is demonstrated.
- ☐ Student explains build versus flash versus run evidence.
- ☐ Code remains inside protected user-code regions.

Marks (10): pre-lab quiz + entry ticket 2 · Blinky 3 · debugger evidence 1 · modification 2 · oral check 2.

Individual oral check

Be prepared for one question and one live change:

- What does a successful build prove?
- What does ST-LINK do?
- Why does the breakpoint pause before or after a particular statement?
- Where should application code be placed, and why?
- What evidence proves your modified firmware is running?

Submit

Submit the requested source/project, one debugger screenshot, the completed prediction table, and a two-sentence debugging report if a failure occurred.

If you are stuck

Report expected and observed behavior; read the **first relevant** error; confirm the project, board, ST-LINK, build, and flash; test whether a breakpoint reaches the loop; then show the TA your evidence and smallest next test.